

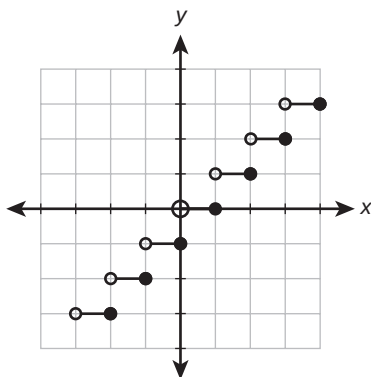
Step Functions

One type of piecewise function is the step function.

Definition

A **step function** is a function defined using a rule that produces a constant value for each designated interval of the domain.

As the name implies, the graph of a step function looks like a set of stairs. Each step is just a line segment with an open or a closed circle on each end. Where the segments are placed and how long they are, as well as whether each endpoint is open or closed, depends on the particular function.



The Greatest Integer Function

The **greatest integer function**, denoted by $f(x) = \lfloor x \rfloor$, assigns the greatest integer less than or equal to each real number in an interval.

► Think About It

NOTATION The greatest integer function may also be given by $f(x) = \lfloor x \rfloor$ or $f(x) = \text{int}(x)$. It is sometimes called the floor function.

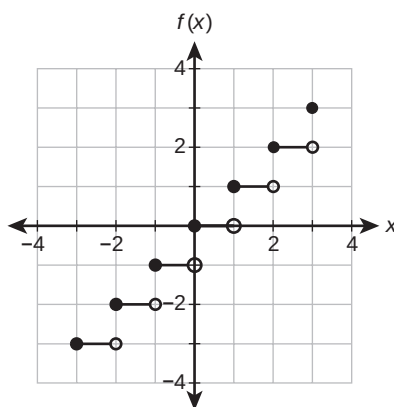
EXAMPLE 1

Graph the function $f(x) = \lfloor x \rfloor$ over the interval $-3 \leq x \leq 3$.

SOLUTION

Make a table of values to examine how the function behaves. The function will behave the same way in each unit interval.

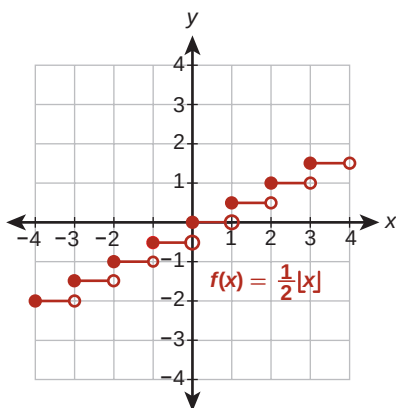
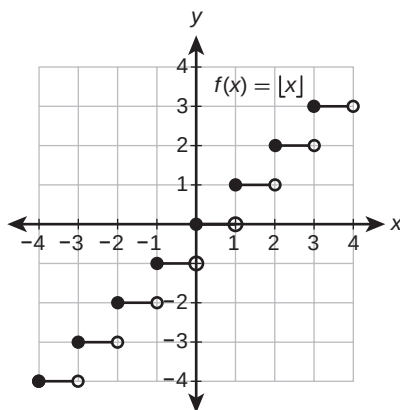
x	$f(x) = \lfloor x \rfloor$
-2.6	-3
-2	-2
-1.3	-2
0	0
0.7	0
1	1
1.2	1
1.5	1
1.9	1
2	2



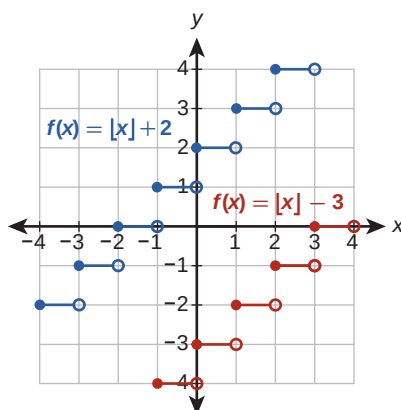
Graphs of Greatest Integer Functions

Greatest Integer Function Graph Family: $f(x) = a \lfloor x \rfloor + k$

Each change in the values of a or k results in a different transformation of the parent graph, $f(x) = \lfloor x \rfloor$.



Changing only the parameter a causes the function to stretch or compress.



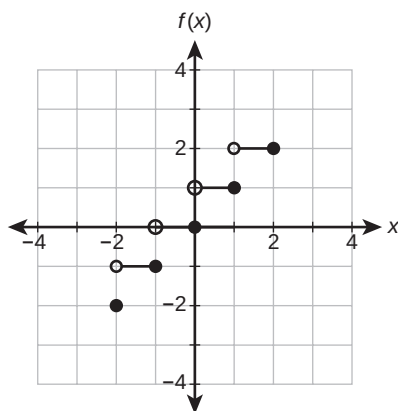
Changing only the parameter k causes the function to shift up or down the y -axis.

The Least Integer Function

The **least integer function**, denoted by $f(x) = \lceil x \rceil$, assigns the least integer greater than or equal to each real number in an interval.

EXAMPLE 2

Write a rule for $f(x) = \lceil x \rceil$ over the interval $-2 \leq x \leq 2$.



SOLUTION

There are four segments. The left endpoint of each segment is graphed with an open circle; the right endpoint is graphed with a solid circle.

To every real number in the interval $(-2, -1]$, the function assigns the value -1 . The rule for this piece is $f(x) = -1$. That is to say, -1 is the least integer greater than or equal to every number in the interval $(-2, -1]$.

x	-2	-1.8	-1	-0.3	0	0.5	1	1.6	2
$f(x) = \lceil x \rceil$	-2	-1	-1	0	0	1	1	2	2

The rule for the function is

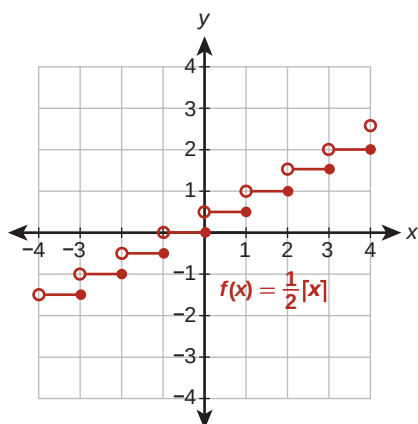
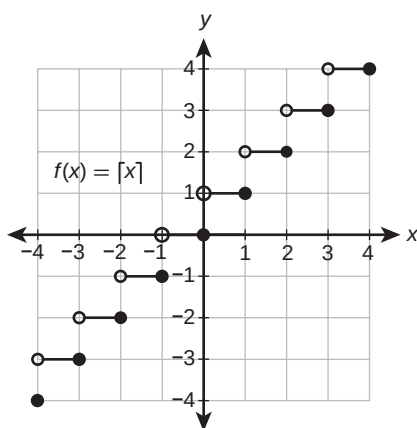
$$f(x) = \begin{cases} -2 & \text{if } x = -2 \\ -1 & \text{if } -2 < x \leq -1 \\ 0 & \text{if } -1 < x \leq 0 \\ 1 & \text{if } 0 < x \leq 1 \\ 2 & \text{if } 1 < x \leq 2 \end{cases}$$

► **Think About It** The least integer function is sometimes called the ceiling function.

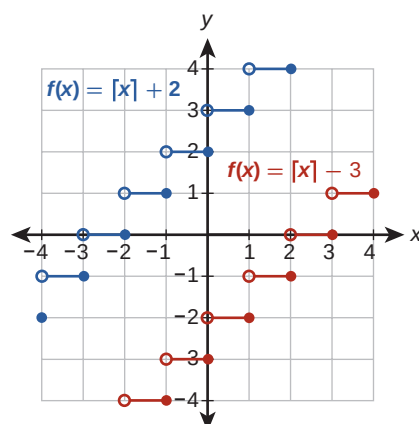
Graphs of Least Integer Functions

Least Integer Function Graph Family: $f(x) = a \lceil x \rceil + k$

Each change in the values of a or k results in a different transformation of the parent graph, $f(x) = \lceil x \rceil$.



Changing only the parameter a causes the function to stretch or compress.



Changing only the parameter k causes the function to shift up or down on the x-axis.

Analyzing the Graph of the Nearest Integer Function

The **nearest integer function**, denoted by $\text{nint}(x)$, assigns the nearest integer to each real number in an interval. To avoid confusion for numbers such as -1.5 and 3.5 , the function assigns the nearest even integer to each input value. So $\text{nint}(-1.5) = -2$, while $\text{nint}(3.5) = 4$. The nearest integer for numbers in the interval $[-0.5, 0.5]$ is zero.

► **Think About It** The nearest integer function is sometimes called the round function.

EXAMPLE 3

Graph the function $f(x) = \text{nint}(x)$ over the interval $-2.5 \leq x \leq 2.5$. Explain the pattern of horizontal segments on the graph.

SOLUTION

Study a table of values of the function for x in the interval $-2.5 \leq x \leq -1.5$. The value of the function for every real number in this interval, including the endpoints, is -2 . The horizontal segment on the graph for this interval will include the endpoints -2.5 and -1.5 .

x	$f(x) = \text{nint}(x)$
-2.5	-2
-2.3	-2
-2.0	-2
-1.8	-2
-1.5	-2

Next, study a table of values of the function for x in the interval $-1.5 \leq x \leq -0.5$. The values assigned to the endpoints of this interval are not the same as the value assigned to the other numbers in the interval. Therefore, the horizontal segment on the graph for this interval will not include the endpoints -1.5 and -0.5 .

x	$f(x) = \text{nint}(x)$
-1.5	-2
-1.3	-1
-1.0	-1
-0.8	-1
-0.5	0

On the graph of the nearest integer function, the segments alternate between those including endpoints (solid circles) and those excluding the endpoints (open circles). This alternating pattern is a result of the rule to round each number to the nearest even integer.

